

10. Create a view to display the fields BookID and BookName where the Publisher is Himalaya.

Q#3 Normalize the following Relation upto Boyce Codd Normal Form (20 Marks)

CUSTOMER-ORDER Table

Invoice ID	InvoiceDate	CustID	CustName	ItemID	ItemName	ItemQty	ItemPrice	Invoice Total
601	14/06/09	C 101	Naveed	1	Scanner	50	\$10	\$2500
				2	Mouse	100	\$20	
				3	Printer	40	\$10	\$400
602	14/06/09	C 101	Naveed	2	Mouse	50	\$20	\$1800
603	14/06/09	C 102	Amjad	4	Mointer	40	\$10	
				1	Scanner	40	\$10	

tep 1: Understanding the Table Structure

Given Table Structure:

- **InvoiceID:** Unique identifier for each invoice.
- **InvoiceDate:** Date when the invoice was issued.
- **CustID:** Unique identifier for each customer.
- **CustName:** Name of the customer.
- **ItemID:** Unique identifier for each item.
- **ItemName:** Name of the item.
- **ItemQty:** Quantity of the item ordered.
- **ItemPrice:** Price of one unit of the item.
- **InvoiceTotal:** Total amount for the invoice.

Step 2: First Normal Form (1NF)

Conditions for 1NF:

1. The table should have no repeating groups or arrays.
2. Each cell must contain a single value (atomic value).

Analysis:

- The table is already in 1NF because each cell contains atomic values.

Step 3: Second Normal Form (2NF)

Conditions for 2NF:

1. The table must be in 1NF.

2. There should be no partial dependency, i.e., non-prime attributes must depend on the entire primary key.

Analysis:

- Here, the composite key could be **(InvoiceID, ItemID)**, as one invoice can contain multiple items.
- **CustName** is dependent on **CustID** alone, and **InvoiceTotal** is dependent on **InvoiceID** alone, which violates 2NF.

Decomposition into 2NF:

1. **CUSTOMER-INVOICE Table:**

- **InvoiceID, InvoiceDate, CustID, CustName, InvoiceTotal**

InvoiceID	InvoiceDate	CustID	CustName	InvoiceTotal
601	14/06/09	C101	Naveed	2500
602	14/06/09	C101	Naveed	400
603	14/06/09	C102	Amjad	1800

ORDER-DETAILS Table:

- **InvoiceID, ItemID, ItemName, ItemQty, ItemPrice**

InvoiceID	ItemID	ItemName	ItemQty	ItemPrice
601	1	Scanner	50	10
601	2	Mouse	100	20
602	3	Printer	40	10
603	2	Mouse	50	20
603	4	Mointer	40	10
603	1	Scanner	40	10

Step 4: Third Normal Form (3NF)

Conditions for 3NF:

1. The table must be in 2NF.

2. There should be no transitive dependency, i.e., non-prime attributes must not depend on other non-prime attributes.

Analysis:

- The **CUSTOMER-INVOICE** table has a transitive dependency since **CustName** depends on **CustID**. To remove this, we need to split it further.

Decomposition into 3NF:

1. **INVOICE Table:**

- **InvoiceID, InvoiceDate, CustID, InvoiceTotal**

InvoiceID	InvoiceDate	CustID	InvoiceTotal
601	14/06/09	C101	2500
602	14/06/09	C101	400
603	14/06/09	C102	1800

2. **CUSTOMER Table:**

- **CustID, CustName**

CustID	CustName
C101	Naveed
C102	Amjad

1. **ORDER-DETAILS Table (remains unchanged):**

- **InvoiceID, ItemID, ItemName, ItemQty, ItemPrice**

Step 5: Boyce-Codd Normal Form (BCNF)

Conditions for BCNF:

1. The table must be in 3NF.
2. For every functional dependency, the left side must be a super key.

Analysis:

- All the tables now follow BCNF, as there are no partial or transitive dependencies, and each left side of the dependency is a super key.

Final Normalized Tables

1. **INVOICE Table:**

- InvoiceID, InvoiceDate, CustID, InvoiceTotal

2. **CUSTOMER Table:**

- CustID, CustName

3. **ORDER-DETAILS Table:**

- InvoiceID, ItemID, ItemName, ItemQty, ItemPrice